

19b 6
part (2)

* PCR

↳ Amplifying specific fragments
of DNA

(polymerase chain rxn)
↳ it can be thought of as Molecular
photocopier.

Kary B. Mullis invented PCR in 1993 (Nobel prize in
chemistry)

PCR machine → Thermal cycler.

* PCR can amplify usable amount of DNA in ~2hr

* PCR can amplify single DNA molecule.

* PCR product can be digested with restriction enzymes,
sequenced or cloned

* PCR Cycling (30-35) cycles

time

each comprising :
for 2nd cycle only
1) Denaturation (95°) 30-60 sec
2) Annealing ($55-65^{\circ}$) 30-60 sec
3) Extension (72°) depends on size



PCR-tube contains:

- ① G-DNA ② buffer ③ Co-factor ($MgCl_2$)
- ④ dNTPs ⑤ F/R primers
- ⑥ DNA polymerase

Forward primer

5' TCGT GGAATAGCA 3' C

3' AGCACCTTATCGTAGC ACAGCTTGGA 5'

5' TCGTGG AATAGCATCGTGTC GAACCTT 3'

3' CAGCTTGGAA 5'

reverse primer

از سمت راست به چپ

* F. primer ? 5' TCGT GGAATAGCA 3'

پیش رو

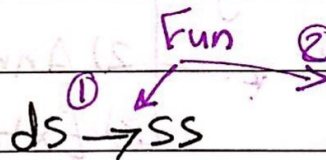
* R. primer ? 5' AAGGTTGAC 3'

پس رو

* PCR rxn :-

→ 1st cycle :

① Initial denaturation



break down H.b

tem	time	# of cycle
90-97	5-10 min	1

	temp	time	# cycle
→ 2nd cycle (three steps)			
① Denaturation	90-97°C	30 sec	30-35
② Annealing	$T_m - 5^\circ\text{C}$	30 sec	
	72°C	depends on size	
→ 3rd cycle			
① Final extension	72°C	5-10 min	
SS → ds			
→ optional cycle	4°C	∞	
* # molecules = $2^n \approx$ # P. cycle			

* Taq polymerase : *Thermus aquaticus*
 Heat stable $\approx 100^\circ\text{C}$
 optimum 72°C

* primer $\text{length} \approx 8$

- 1) 100% complementary template
- 2) length (18-24 bp)
- 3) GC content (GC% : 40-60%)
- 4) T_m difference $< 5^\circ\text{C}$
- 5) Annealing ~~temp~~ ($T_m - 5^\circ\text{C}$)
 $\rightarrow$ المدة
- b) Dimering $\leq 90\%$

(primer $\rightarrow$ 18-24 bp)
 (F. PR $\rightarrow$ length)
 $\rightarrow$ الطول

Lab 7

* Allele specific PCR

SNP or mutation

5' GATCACC GAT CGAAT TACGATAT 3'

F primer: 5' GATCACC GATCGAAT

F primer: 5' GATCACC GATCGAAC

R primer: 5' ATATCGTA.....

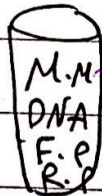
* SNP → $\text{iso} \rightarrow \text{primer 1}$

* the most important nucleotide in primer is the one at 3'

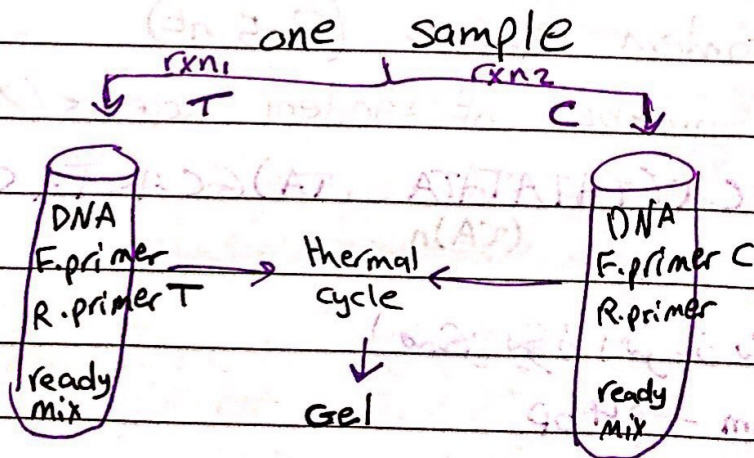
non specific → ampli — nucl primer 1

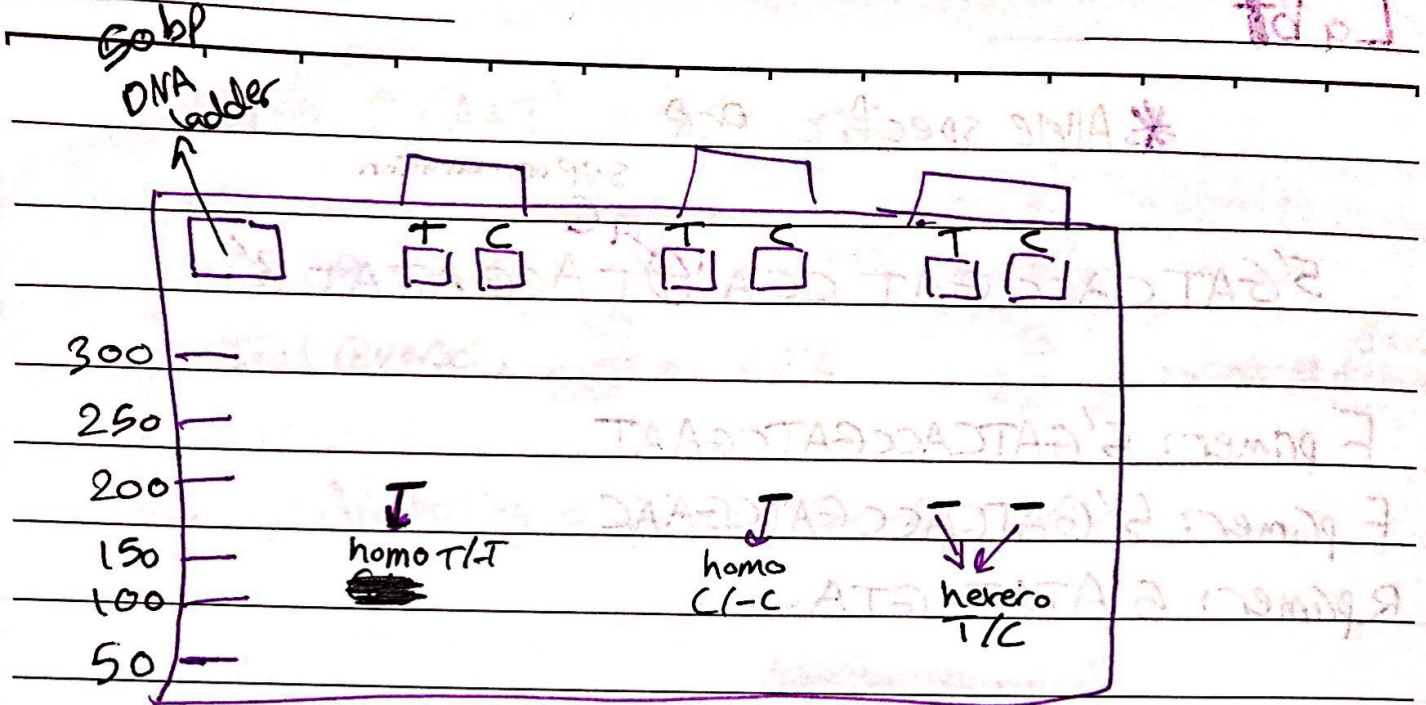
ampli — primer 1 nucl

Traditional PCR



ready mix





2% Agarose gel

Forward و reverse ال SNP ال
band ال Homo ال
T C
homo A-A

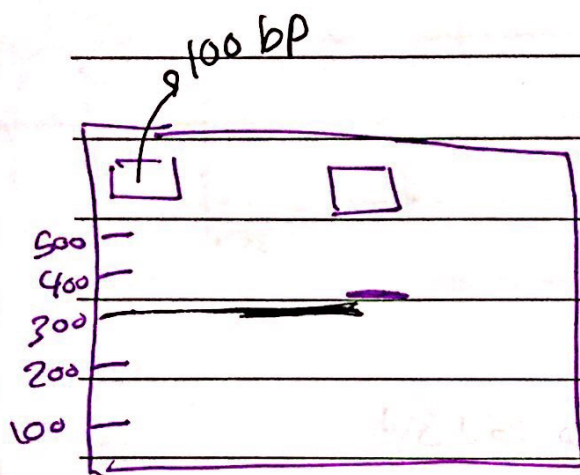
* STR → Short tandem repeats (2-5 nt)

* VNTR → Variable number of tandem Repeats (≥ 10 nt)

5' CTGAGGCT (TATATATA...TA) GCACTAC 3'
(TA)_n

(ممكن يكون اكثر من 10 تكرار)

Maximum → 4 bp



F. primer 18 bp

R. Primer 18 bp

$$\text{PCR-Product} = 300 \text{ bp} - (18 + 18)$$

$$= 300 - 36$$

$$= 264 = 132 \text{ repeat}$$

عدد ال repeat unit

repeat unit

* Primers

stock (1)

$$M = \frac{\# \text{ moles}}{\text{Volume}}$$

$$200 = \frac{79.9 \times 10^3}{V}$$

$$V = 399.5 \mu\text{l} \text{ --- NFW}$$

↓
vortex

↓

2hr RT

↓

Freezer 0°C

(UV hood لقياس كمية DNA)

* Working Prime :

$$C_1 V_1 = C_2 V_2$$

$$200 \times V_1 = 10 \text{ p.moles} \times 1000 \mu\text{l}$$

$$V_1 = \frac{\text{stock}}{\text{NFW}} = 50 \mu\text{l} + 950 \mu\text{l}$$

$$= 1000 \mu\text{l} = 10 \text{ p.mole}$$

* PCR rxn for 1 sample :

Ready mix	12.5
F. primer	1.3
R. primer	1.3
N.F.W	7.9 منونة الباقي لـ 25
<u>DNA</u>	2 μ l
Final volume	25 μ l معطيات

- Ready Mix : $C_1 V_1 = C_2 V_2$

$$(2x) V_1 = (1x) 25$$

$$V_1 = 12.5 \mu\text{l}$$

- R, F primer : $C_1 V_1 = C_2 V_2$

$$10 V_1 = 0.5 \times 25$$

$$V_1 = 1.25 \approx 1.3$$

* PCR for 2 samples :

هون بنمیزب عربا الفیات + 1

$$0 = 1 + 5$$

R.M $\rightarrow 12.5 \times 5 = 62.5$

F. primer $\rightarrow 1.3 \times 5 = 6.5$

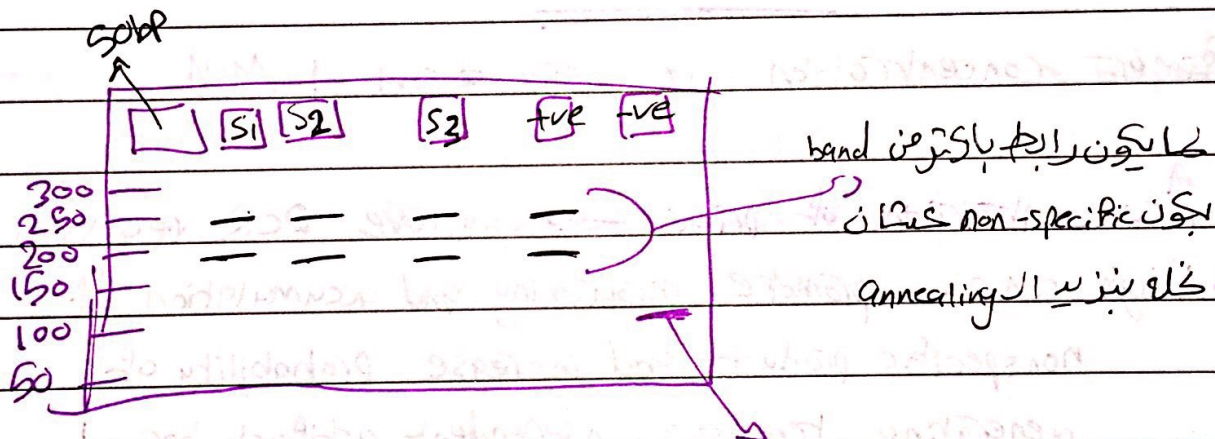
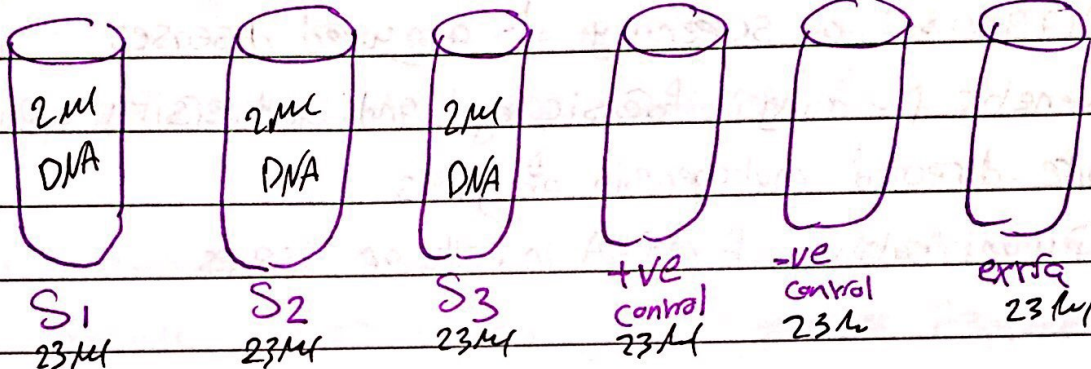
R. primer $\rightarrow 1.3 \times 5 = 6.5$

NFW $\rightarrow 7.9 \times 5 = 39.5$

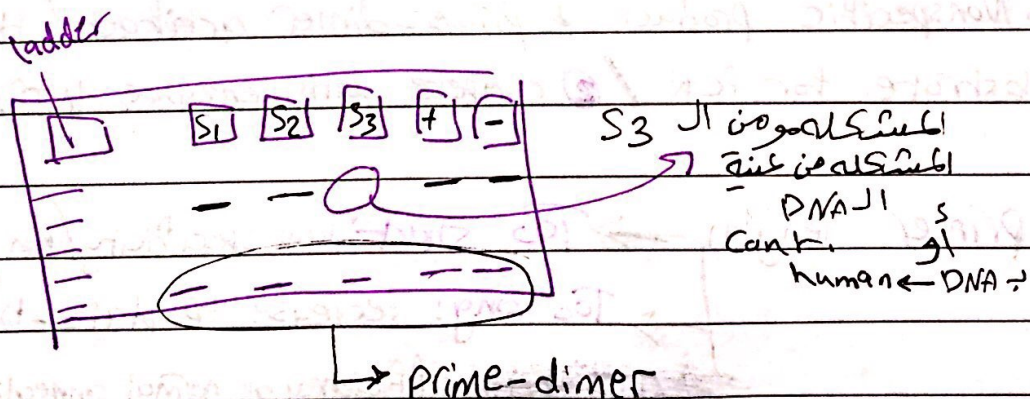
DNA $\rightarrow 2 \mu\text{l}$

PKD → 46 exons

pendrop



الى يكون اقل من الفيات يكون
Con. bacteria/virus



والهنا نرى / سيجو دورهم : زيادة Conc. ال primer
تظلم منها كما تقل ال Conc.

* PCR Applications :-

- Mutation testing e.g: cystic Fibrosis
- SNP genotyping
- Diagnosis or screening of acquired diseases
- Genetic profiling in forensic, legal and biodiversity application
- Site-directed mutagenesis of genes
- Quantitation of mRNA in cells or tissues

* Primer concentration in PCR = 0.1 - 1 μ M

* $\uparrow$ concentration of primer $\rightarrow$ improve PCR reaction

* High conc. promote mispriming and accumulation of nonspecific product and increase probability of generating template-independent artifact termed

primer-dimer

* Nonspecific product & primer-dimer artifact ^{① are} themselves substrate for PCR / ② compete with enzyme, dNTPs, primers,

* Primer length $\rightarrow$ Too short: low specificity / non specific

$\rightarrow$ Too long: decrease template-binding

~~efficiency~~ efficiency at normal annealing temp due to high probability to form hairpins

(Lower T_m)

- * Optimal length :
 - 18-24bp for general application
 - 30-35 bp for multiplex PCR

- * Optimal amplicon size :
 - 300-1000 bp for general application, avoid $>3\text{Kb}$
 - 50-150 bp for RT PCR, avoid $>400\text{bp}$

- * Cross homology $\rightarrow$ become problem when PCR template is genomic DNA or mixed gene ~~fragments~~ fragments

- * GC clamp $\rightarrow$ presence of G or C within last 4 base from 3' end of primer (essential for preventing mis priming & enhancing specific primer template binding).

* Primer pair T_m mismatch:
Significant mismatch can lead to poor amplification.
Desirable T_m difference $< 5^\circ\text{C}$ between primer pair.

* Optimal melting temp between $52^\circ\text{C} - 65^\circ\text{C}$
- above 65°C should avoid because of potential
for secondary annealing
- higher T_m ($75^\circ\text{C} - 80^\circ\text{C}$) recommended for amplifying
high GC content targets

* long runs increase non specific annealing
- Maximum acceptable num of runs $\rightarrow 4\text{bp}$